



UCD College of Business

UCD College of Business, Global Campus

Assessment Submission Form

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Date: 26/12/2024

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Group Assignment Solution: Investment Analysis

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1 Data Collection and Processing

This report investigates two stocks using the closing prices for Apple Inc. (AAPL) and Coca-Cola Co. (KO) from Yahoo Finance from January 2019 to December 2023. The data will be used to carry out a portfolio analysis of the two assets, and the findings will be used to offer investment recommendations to potential investors.

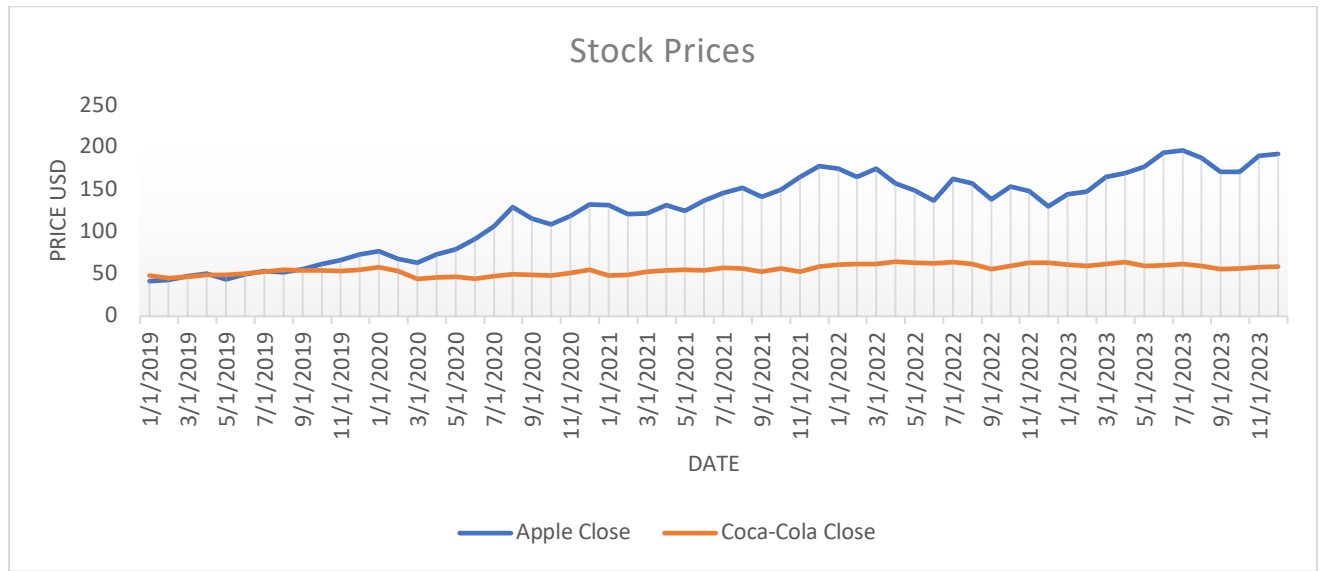


Figure 1 Stock Prices

Based on the distribution of the prices for both stocks, Apple stock prices are more volatile than Coca-Cola's. The following formula was used to calculate the monthly returns of each asset.

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100$$

In this state, return R_t is the monthly returns

- P_t represents the closing price at month t
- P_{t-1} is the closing price at month $t-1$.

Table 1 Monthly stock returns Statistics.

Coca-Cola Monthly Return		Apple Monthly Return	
Mean	0.004902166	Mean	0.029961
Standard Error	0.007010016	Standard Error	0.011321
Median	0.0172151	Median	0.044613
Mode	#N/A	Mode	#N/A
Standard Deviation	0.053844952	Standard Deviation	0.086958
Sample Variance	0.002899279	Sample Variance	0.007562
Kurtosis	1.250778026	Kurtosis	-0.88716
Skewness	-0.798485289	Skewness	-0.14051
Range	0.301627213	Range	0.341952
Minimum	-0.172742595	Minimum	-0.12757
Maximum	0.128884618	Maximum	0.21438
Sum	0.289227805	Sum	1.767712
Count	59	Count	59

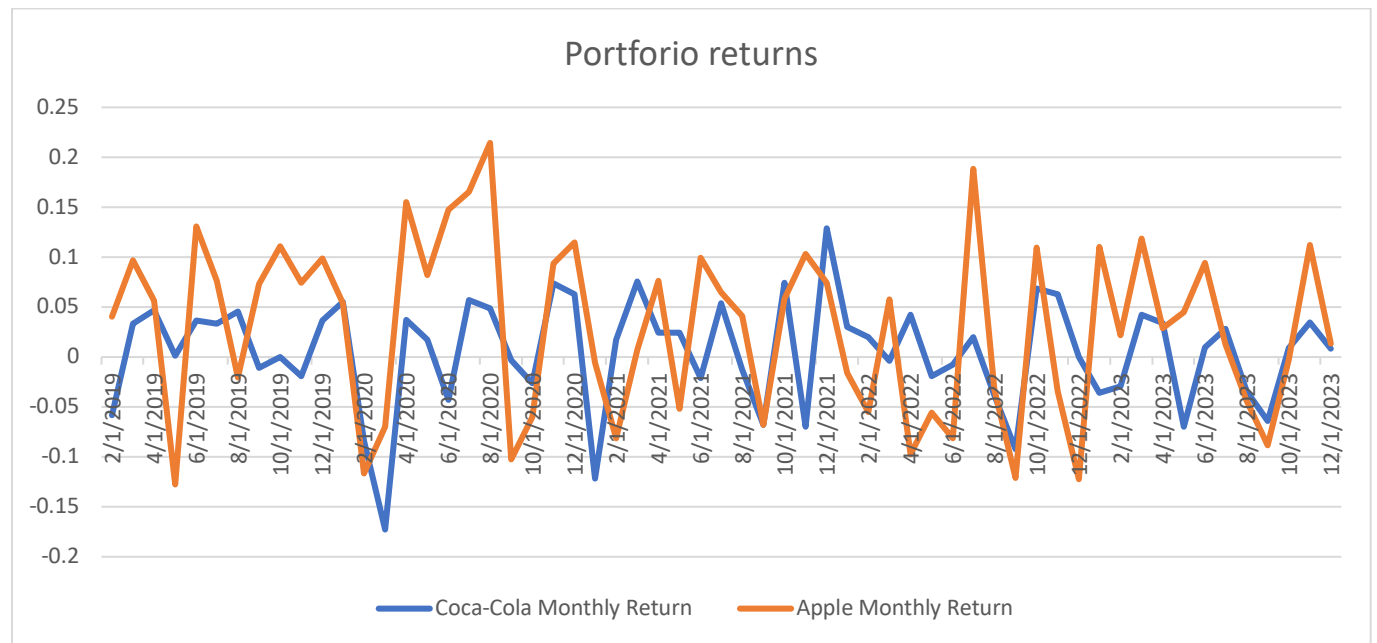


Figure 2 Distribution of Monthly returns

Based on the monthly returns for the five years, Apple's returns are more volatile than those of Coca-Cola. This observation may indicate that although Apple may have had higher returns, it is

associated with a higher risk than Coca-Cola. For data cleaning, missing observations generated during the creation of the returns on the assets were dropped. The returns had no outliers, and all the observations were within a specific range for both stocks.

Annualised Mean

$$AR = \left((1 + r_{\text{monthly}})^{12} - 1 \right) \times 100$$

(r_{monthly}) is the average monthly return.

12 represents the number of months in a year.

Coca-Cola:

$$AR_{\text{Coca-Cola}} = ((1 + 0.004902166)^{12} - 1) \times 100$$

$$1 + 0.004902166 = 1.004902166$$

$$1.004902166^{12} = 1.060438261$$

$$1.060438261 - 1 = 0.060438261$$

$$0.060438261 \times 100 = 6.04\%$$

Apple:

$$AR_{\text{Apple}} = ((1 + 0.029961222)^{12} - 1) \times 100$$

$$1 + 0.029961222 = 1.029961222$$

$$1.029961222^{12} = 1.42511688$$

$$1.42511688 - 1 = 0.42511688$$

$$0.42511688 \times 100 = 42.51\%$$

Annualized Standard deviation

$$\text{Annualized Std Dev} = \text{Monthly Std Dev} \times \sqrt{12}$$

Coca-Cola:

$$\text{Monthly Std Dev}_{\text{Coca-Cola}} = 0.053844952$$

The annualized standard deviation is:

$$\text{Annualized Std Dev}_{\text{Coca-Cola}} = 0.053844952 \times \sqrt{12} = 0.1863$$

Apple:

$$\text{Monthly Std Dev}_{\text{Apple}} = 0.086958472$$

$$\text{Annualized Std Dev}_{\text{Apple}} = 0.086958472 \times \sqrt{12} = 0.3023$$

1.1 Correlation

The Excel data analysis tool was used to generate the Pearson correlation coefficient for the correlation.

Table 2 Correlation Between Stock Returns

	<i>Coca-Cola Monthly Return</i>	<i>Apple Monthly Return</i>
Coca-Cola Monthly Return	1	
Apple Monthly Return	0.385316106	1

The correlation coefficient for the monthly returns of the two stocks is 0.3853, indicating a moderate positive correlation between the two stock returns. This observation shows that increased Coca-Cola returns are expected to increase Apple's monthly returns by 0.3853 and vice versa. It is important to note that correlation does not imply causation, and other factors not included in this analysis may affect the positive change in the assets (Pilbeam, 2005), which may not be linked directly to the two stock returns

2 Investment Opportunity Set

We analyse the relationship between portfolio returns and risks by varying the weight of Coca-Cola in the portfolio from 0% to 150% in increments of 5%. The primary focus is to examine the trade-off between risk and return for different investment combinations of the two stocks.

2.1 Portfolio Return

The portfolio return is calculated as a weighted average of the individual returns of Coca-Cola and Apple. The weight of Coca-Cola in the portfolio is denoted by (w_1), and the weight of Apple is ($1 - w_1$).

$$r_p = w_1 \cdot r_{\text{Coca-Cola}} + (1 - w_1) \cdot r_{\text{Apple}}$$

($r_{\text{Coca-Cola}}$) is the return of Coca-Cola

(r_{Apple}) is the return of Apple.

2.2 Portfolio Risk

The portfolio risk (standard deviation) is calculated using the formula for the combined risk of two assets, considering their standard deviations and correlation:

$$\sigma_p = \sqrt{w_1^2 \cdot \sigma_{\text{Coca-Cola}}^2 + (1 - w_1)^2 \cdot \sigma_{\text{Apple}}^2 + 2 \cdot w_1 \cdot (1 - w_1) \cdot \text{Cov}(r_{\text{Coca-Cola}}, r_{\text{Apple}})}$$

The covariance is calculated as:

$$\text{Cov}(r_{\text{Coca-Cola}}, r_{\text{Apple}}) = \rho \cdot \sigma_{\text{Coca-Cola}} \cdot \sigma_{\text{Apple}}$$

with (ρ) being the correlation between Coca-Cola and Apple.

Coca-Cola Monthly Return: ($r_{\text{Coca-Cola}} = 0.004902166$)

Apple Monthly Return: ($r_{\text{Apple}} = 0.029961222$)

Coca-Cola Standard Deviation: ($\sigma_{\text{Coca-Cola}} = 0.053844952$)

Apple Standard Deviation: ($\sigma_{\text{Apple}} = 0.086958472$)

Correlation between Coca-Cola and Apple: ($\rho = 0.385316106$)

Table 3 Returns and Risk Trade-off

Weights of Coca-Cola (w1)	Portfolio Return	Portfolio StdDev
0.00	0.0300	0.0870
0.05	0.0287	0.0837
0.10	0.0275	0.0805
0.15	0.0262	0.0774
0.20	0.0249	0.0744
0.25	0.0237	0.0715
0.30	0.0224	0.0687
0.35	0.0212	0.0661
0.40	0.0199	0.0637
0.45	0.0187	0.0614
0.50	0.0174	0.0593
0.55	0.0162	0.0575
0.60	0.0149	0.0559
0.65	0.0137	0.0545
0.70	0.0124	0.0535
0.75	0.0112	0.0527
0.80	0.0099	0.0523
0.85	0.0087	0.0522
0.90	0.0074	0.0524
0.95	0.0062	0.0530
1.00	0.0049	0.0538
1.05	0.0036	0.0550
1.10	0.0024	0.0565
1.15	0.0011	0.0582
1.20	-0.0001	0.0601
1.25	-0.0014	0.0623

1.30	-0.0026	0.0646
1.35	-0.0039	0.0671
1.40	-0.0051	0.0698
1.45	-0.0064	0.0726
1.50	-0.0076	0.0755

Based on the expected returns, a higher investment in Apple results in higher returns than in Coca-Cola. There is, therefore, a need to investigate the optimum investment allocation in the two assets that give the highest returns but simultaneously have the lowest expected risk.

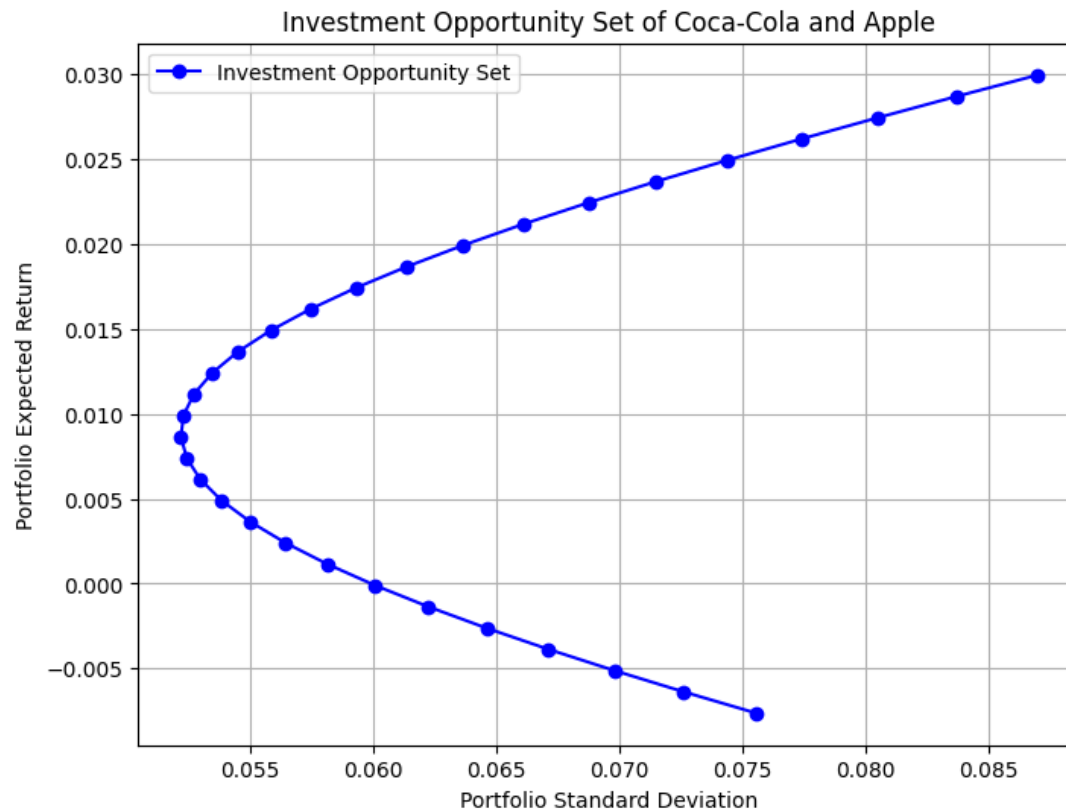


Figure 3 Investment Opportunity

3 Optimal Risky Portfolio (ORP) Calculation

The weights of stock 1 (Coca-Cola) and stock 2 (Apple) in the optimal risky portfolio (ORP) are calculated using the following formulae. The risk-free rate for this calculation was assumed to be 0.0002,

The following formula gives the weights for the optimal risky portfolio:

$$w_1 = \frac{[E(r_1) - r_f]\sigma_2^2 - [E(r_2) - r_f]\rho_{12}\sigma_1\sigma_2}{[E(r_1) - r_f]\sigma_2^2 + [E(r_2) - r_f]\sigma_1^2 - ([E(r_1) - r_f] + [E(r_2) - r_f])\rho_{12}\sigma_1\sigma_2}$$

$$w_2 = 1 - w_1$$

$$[E(r_1) = 0.004902, \quad E(r_2) = 0.029961, \quad r_f = 0.0002]$$

Standard deviations: $[\sigma_1 = 0.053845, \quad \sigma_2 = 0.086958]$

Correlation coefficient: $[\rho_{12} = 0.3853]$

Differences from Risk-Free Rate

$$E(r_1) - r_f = 0.004902 - 0.0002 = 0.004702$$

$$E(r_2) - r_f = 0.029961 - 0.0002 = 0.029761$$

$$w_1 = \frac{(0.004702)(0.086958^2) - (0.029761)(0.3853)(0.053845)(0.086958)}{(0.004702)(0.086958^2) + (0.029761)(0.053845^2) - ((0.004702) + (0.029761))(0.3853)(0.053845)(0.086958)}$$

$$w_1 = -0.3040$$

$$w_2 = 1.3040$$

$$E(R_p) = w_1 \cdot E(R_1) + w_2 \cdot E(R_2)$$

$$E(R_p) = (-0.3040) \cdot 0.004902 + (1.3040) \cdot 0.029961$$

$$E(R_p) = 0.0376$$

The standard deviation of the ORP is calculated as:

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2}$$

σ_p

$$= \sqrt{(-0.3040)^2 \cdot 0.053845^2 + (1.3040)^2 \cdot 0.086958^2 + 2 \cdot (-0.3040) \cdot (1.3040) \cdot 0.3853 \cdot 0.053845 \cdot 0.086958}$$

$$\sigma_p = 0.1081$$

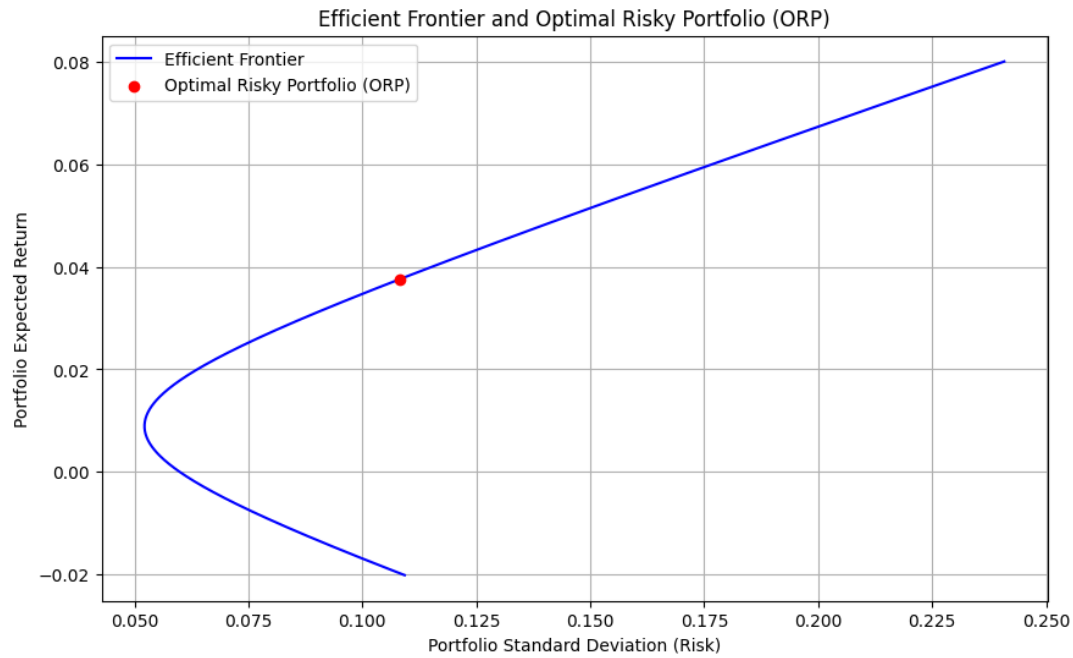


Figure 4 Graph of Investment Opportunity Set

4 Minimum Variance Portfolio (MVP)

To calculate the weights for the Minimum Variance Portfolio (MVP), we use the following formulas:

$$w_1 = \frac{\sigma_2^2 - \rho_{12} \sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12} \sigma_1 \sigma_2}$$

$$w_2 = 1 - w_1$$

Where:

–(w_1) is the weight of Stock 1 (Coca-Cola),

–(w_2) is the weight of Stock 2 (Apple),

–(σ_1) is the standard deviation of Stock 1 (Coca-Cola),

–(σ_2) is the standard deviation of Stock 2 (Apple),

–(ρ_{12}) is the correlation between Stock 1 and Stock 2.

$$\sigma_1 = 0.053845, \quad \sigma_2 = 0.086958, \quad \rho_{12} = 0.3853$$

$$w_1^{\text{MVP}} = \frac{(0.086958)^2 - (0.3853) \times (0.053845) \times (0.086958)}{(0.053845)^2 + (0.086958)^2 - 2 \times (0.3853) \times (0.053845) \times (0.086958)}$$

$$w_1^{\text{MVP}} = \frac{0.0075646 - 0.0015671}{0.0028957 + 0.0075646 - 0.0029222} = 0.8402$$

$$w_1^{\text{MVP}} = 0.8402$$

$$w_2^{\text{MVP}} = 1 - 0.8402 = 0.1598$$

$$w_1^{\text{MVP}} = 0.8402$$

$$w_2^{\text{MVP}} = 0.1598$$

5 Expected Return and Standard Deviation of the MVP

The weight of Stock 1 (Coca-Cola) in the Minimum Variance Portfolio (MVP) is given by the formula:

$$w_1^{\text{MVP}} = \frac{\sigma_2^2 - \rho_{12}\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}$$

Now, the expected return of the Minimum Variance Portfolio (MVP) is calculated as a weighted average of the individual expected returns:

$$E(R_{\text{MVP}}) = w_1^{\text{MVP}} \cdot E(R_1) + w_2^{\text{MVP}} \cdot E(R_2)$$

Where:

$$-(E(R_1) = 0.004902) \text{ (Expected return of Coca-Cola)}$$

$$-(E(R_2) = 0.029961) \text{ (Expected return of Apple)}$$

$$E(R_{\text{MVP}}) = (0.8402)(0.004902) + (0.1598)(0.029961)$$

$$E(R_{\text{MVP}}) = 0.004115 + 0.004787 = 0.008902$$

$$E(R_{\text{MVP}}) = 0.0089$$

Finally, the standard deviation of the Minimum Variance Portfolio (MVP)

$$\sigma_{\text{MVP}} = \sqrt{w_1^{\text{MVP}^2} \sigma_1^2 + w_2^{\text{MVP}^2} \sigma_2^2 + 2w_1^{\text{MVP}} w_2^{\text{MVP}} \rho_{12} \sigma_1 \sigma_2}$$

$$\sigma_{\text{MVP}}$$

$$= \sqrt{(0.8402)^2 (0.053845)^2 + (0.1598)^2 (0.086958)^2 + 2(0.8402)(0.1598)(0.3853)(0.053845)(0.086958)}$$

$$\sigma_{\text{MVP}} = \sqrt{0.0002434 + 0.0002389 + 0.0000704} = \sqrt{0.0005527}$$

$$\sigma_{\text{MVP}} = 0.0522$$

Thus, the standard deviation of the MVP is:

$$\sigma_{\text{MVP}} = 0.0522$$

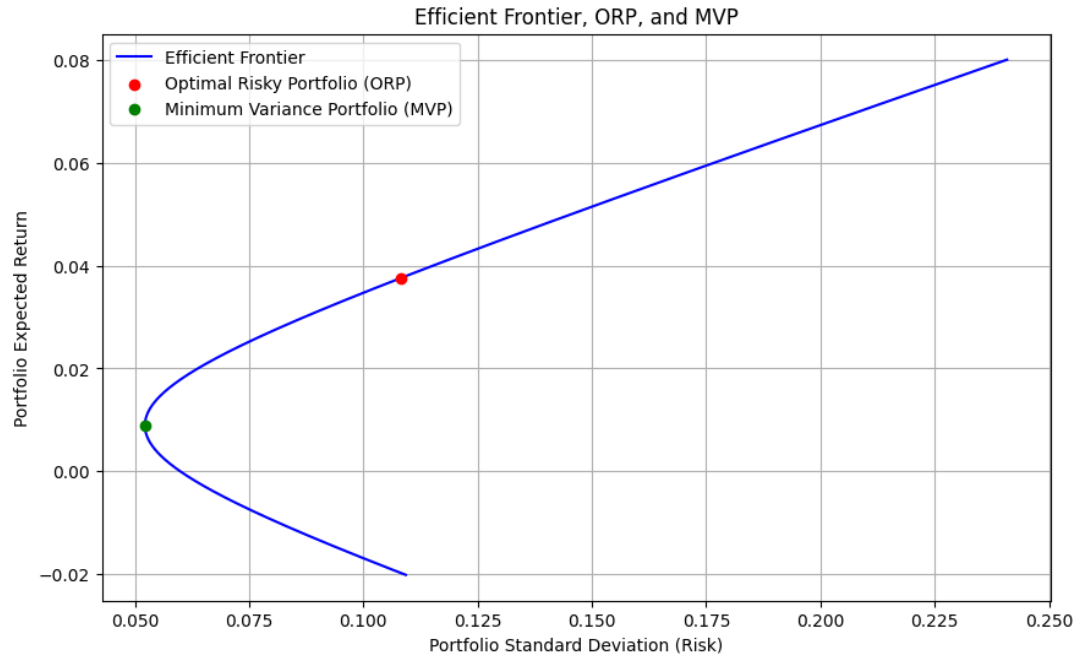


Figure 5 *Efficient Frontier*

6 Discussion

This section aims to report on the findings of the analysis of diversification. In asset investment, diversification is mainly employed to reduce the expected risk in a portfolio and maximize the expected returns such that the investor maximizes the expected returns. Diversification may be more applicable in assets that don't have a high correlation as a higher correlation may mean that the two assets may share both the risk and returns; hence, it is not a good way of diversification (Koumou, 2020). In this analysis, the Optimal Risky Portfolio (ORP) and the Minimum Variance Portfolio (MVP) have been plotted and serve as two critical points on the efficient frontier. The results obtained will help identify and make a recommendation about the tradeoff between risk and return that every investor must consider when investing in Coca-Cola and Apple based on the monthly returns between January 2019 and December 2023. It is important to note that different factors in the market may affect the returns on various assets. Therefore, there is a need to investigate other possible effects on the market performance before

investing. For example, the monthly returns for Apple and Coca-Cola are different throughout the months, as shown in Figure 2. Based on the distribution, higher volatility in returns can be seen in 2020, which decreases in 2021 and is higher again in 2022. The change in 2020 and part of 2021 may have been attributed to the effect of COVID-19 on the stock market(Nian et al., 2021). This observation shows that the returns for different stocks, in this case, Apple and Coca-Cola, may be affected by time and other forces in the market that should also be considered when making a portfolio diversification decision(Jansen & Zervou, 2017). Similarly, uncertainty in the market may cause the stock market returns to turn volatile(Benchimol et al., 2023).

6.1 Annualized Returns and standard deviation

Based on the annualized returns for the two assets, Coca-Cola had an average annual return of 6.04%, while Apple had an average yearly return of 42.51%. The annualized standard deviation for Apple was 0.302, and Coca-Cola's was 0.1863. This observation indicates that Apple may have been more volatile in monthly annualized returns than Coca-Cola. The observation also supports that Apple returns are riskier than Coca-Cola's due to the higher annualized standard deviation. The correlation coefficient between the two assets is 0.385, which indicates a slightly moderate positive correlation. Based on the findings, investing more in Apple may result in higher returns than Coca-Cola, and an investor may choose to invest higher shares in Apple than in Coca-Cola. However, balancing a portfolio may be advisable to prevent risk and losses from one asset compared. Therefore, diversification helps balance the returns between the assets to achieve optimal risk and returns.

6.2 Investment Opportunity

Based on the investment opportunity calculation, the expected portfolio performance was calculated using 0% to 150% increments of 5%. Based on the results, investing less in Coca-Cola led to higher returns than Apple. However, there is a need to identify the optimal investment tradeoff between the two assets. Hence, the use of Optimal risky portfolio and Minimum variance portfolio. The Optimal Risky Portfolio is mainly applied to maximise returns(Mondello, 2023). In this case, an investor focuses

on the best combination of risky assets they should hold, assuming they are not investing in risk-free assets. In the Minimum Variance Portfolio (MVP), the focus is to achieve the lowest risks possible among the assets (Clarke et al., 2010), for this case, the standard deviation. MVP becomes more valuable for risk-averse investors who seek to minimise risk as much as possible. Therefore, MVP may result in slightly lower returns than ORP since lower-risk portfolios are expected to have lower volatility (Russo, 2016), hence a lower margin of returns. Higher risks are always rewarded with higher returns (Chung & Chuwongnant, 2018; Pankunni & Kumar, 2023).

The calculated weights for the two assets in the Minimum Variance Portfolio were (Coca-Cola) 0.8402 and Stock 2 (Apple) 0.1598. Additionally, the expected return of the MVP was 0.0089 (or 0.89%), and the standard deviation was 0.0522 (or 5.22%). On the other hand, the ORP provides a much higher expected return (3.76%) but comes with higher risk (10.81%). The ORP offered a higher expected return of 3.76% at a risk level of 10.81% by allocating -30.40% (short position) to Coca-Cola and 130.40% to Apple. This would mean the investor borrows and sells Coca-Cola stock to free up additional capital (Fabozzi, 2004; Khodamoradi et al., 2021). This observation can graphically be seen in Figure 5, where the efficient frontier shows the ORP to have higher returns than the MVP, the curve shows all optimal portfolios for a given level of risk, and an investor may be able to reference and choose different levels based on their desired risks. Therefore, through the efficient frontier, the ORP may give higher returns if the investor is more risk-averse.

This study investigated the annual returns for two assets, Apple and Coca-Cola. Based on the annualised returns, Apple had a higher return on investment compared to Coca-Cola but, at the same time, a higher risk. In terms of diversifying the portfolio, there was the need for a tradeoff between the risk and the expected returns of both assets as one portfolio. In that case, based on the MVP and ORP, The ORP offered a higher expected return of 3.76% at a risk level of 10.81% by allocating -30.40% (short position) to Coca-Cola and 130.40% to Apple, assuming that the allocation. The assumption is that each asset's

allocation should equal 100% of the total portfolio. The high allocation to Coca-Cola (84.02%) and the lower allocation to Apple (15.98%) helped achieve this reduced MVP risk.

The positive correlation between the returns in the two assets may indicate that better performance in returns in one asset may be associated with positive performance in the other by a 0.385 correlation coefficient. Based on the findings, the higher investment into Apple within the last 5 years from 2023 may have resulted in a higher return than Coca-Cola when all other factors are constant. It is important to note that the calculations assume a 0.02% risk-free rate and that all other market factors are constant. It is important to note that the analysis is developed under various assumptions, and the findings can only hold if the assumptions are maintained(Wilms et al., 2021).

7 Recommendations and Conclusion

The selection of MVP for risk-averse investors offers higher chances of capital preservation as these actions could be well suited for pension funds. The ORP could be the asset class to go for in the case of investors capable of assuming more risk. This action is well suited for long-term investments and may be associated with some losses expected when investing in the short term once the asset is volatile(ATAS, n.d.). A growth investor seeks high returns and can withstand fluctuations in risk that usually occur in the portfolio's value in the short term(Betermier et al., 2017).

Combining ORP and MVP could yield more diversification on their own; this mix combines both elements from ORP and MVP(Jiang et al., 2019). Similarly, the investor should be able to withstand risks associated with economic downturns, especially in an event similar to COVID-19(Angosto-Fernández & Ferrández-Serrano, 2024). This gives an investor more chances to achieve the desired risk-return tradeoff. It would be ideally suited for those investors who combine moderate risk with high-return assets.

Finally, the investor must monitor the macroeconomic factors affecting the performance of a specific portfolio. These include monitoring the performance of a company and the market(Lai, 2017). Most importantly, based on the analysis of the OFP and MVP, the investor will choose between two opposed

approaches to risk-return management. The MVP is recommended to the risk-averse, while the ORP will be an option for those tolerating high risk for the sake of high returns. This gives the investor a clear picture of their investment philosophy, risk preferences, and market condition and makes an informed decision to optimize their portfolio for long-term feasible investment objectives.

Rendering close monitoring to the allocated portfolio through leanness of profits or losses, be it MVP, ORP, or a combined use of both. Adaptive rebalancing will suit a fluctuating period of investments to achieve optimal results.

(2418 Words).

8 References

- Angosto-Fernández, P. L., & Ferrández-Serrano, V. (2024). Market risk exposure determinants during the COVID-19 outbreak: between competitiveness and inequality. *International Journal of Emerging Markets*. <https://doi.org/10.1108/IJOEM-01-2023-0080>
- ATAS. (n.d.). *Volatility: Meaning in Finance and How It Works With Stocks* | ATAS. Retrieved December 20, 2024, from <https://atas.net/market-theory/volatility/>
- Benchimol, J., Saadon, Y., & Segev, N. (2023). Stock market reactions to monetary policy surprises under uncertainty. *International Review of Financial Analysis*, 89. <https://doi.org/10.1016/j.irfa.2023.102783>
- Betermier, S., Calvet, L. E., & Sodini, P. (2017). Who Are the Value and Growth Investors? *Journal of Finance*, 72(1). <https://doi.org/10.1111/jofi.12473>
- Chung, K. H., & Chuwonganant, C. (2018). Market volatility and stock returns: The role of liquidity providers. *Journal of Financial Markets*, 37. <https://doi.org/10.1016/j.finmar.2017.07.002>
- Clarke, R., De Silva, H., & Thorley, S. (2010). Minimum-Variance Portfolio Composition. *Journal of Portfolio Management*, 37(2), 31–45. <https://doi.org/10.3905/JPM.2011.37.2.031>
- Fabozzi, F. J. . (2004). *Short Selling: Strategies, Risks, and Rewards*. 416.
- Jansen, D. W., & Zervou, A. (2017). The time varying effect of monetary policy on stock returns. *Economics Letters*, 160, 54–58. <https://doi.org/10.1016/J.ECONLET.2017.08.022>
- Jiang, C., Du, J., & An, Y. (2019). Combining the minimum-variance and equally-weighted portfolios: Can portfolio performance be improved? *Economic Modelling*, 80, 260–274. <https://doi.org/10.1016/J.ECONMOD.2018.11.012>
- Khodamoradi, T., Salahi, M., & Najafi, A. R. (2021). Cardinality-constrained portfolio optimization with short selling and risk-neutral interest rate. *Decisions in Economics and Finance*, 44(1), 197–214. <https://doi.org/10.1007/S10203-020-00293-9/METRICS>
- Koumou, G. B. (2020). Diversification and portfolio theory: a review. *Financial Markets and Portfolio Management*, 34(3), 267–312. <https://doi.org/10.1007/S11408-020-00352-6/METRICS>

- Lai, Y. W. (2017). Macroeconomic factors and index option returns. *International Review of Economics & Finance*, 48, 452–477. <https://doi.org/10.1016/J.IREF.2016.11.002>
- Mondello, E. (2023). *Optimal Portfolio*. 145–185. https://doi.org/10.1007/978-3-658-41021-6_5
- Nian, R., Xu, Y., Yuan, Q., Feng, C., & Lendasse, A. (2021). Quantifying Time-Frequency Co-movement Impact of COVID-19 on U.S. and China Stock Market Toward Investor Sentiment Index. *Frontiers in Public Health*, 9. <https://doi.org/10.3389/fpubh.2021.727047>
- Pankunni, N., & Kumar, S. R. (2023). Volatility of Returns in Stock Market Investments: A Study of BRICS Nations. *Finance: Theory and Practice*, 27(2). <https://doi.org/10.26794/2587-5671-2023-27-2-87-98>
- Pilbeam, K. (2005). Portfolio Analysis: Risk and Return in Financial Markets. *Finance and Financial Markets*, 156–187. https://doi.org/10.1007/978-1-349-26273-1_7
- Russo, A. (2016). Low-risk equity investment-From theory to practice. *Journal of Asset Management*, 17(4), 264–279. <https://doi.org/10.1057/JAM.2016.13/FIGURES/5>
- Wilms, R., Mäthner, E., Winnen, L., & Lanwehr, R. (2021). Omitted variable bias: A threat to estimating causal relationships. In *Methods in Psychology* (Vol. 5). <https://doi.org/10.1016/j.metip.2021.100075>

9 Academic Integrity





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